

The Selector Principle and Its Dual Witnesses: A Concordance for the Obstruction Programme

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September 24, 2026

Abstract

One object organizes the programme’s certificates: the recursively viable selector $\Gamma_N(B)$, whose emptiness at any factor is the obstruction, whose dual witnesses are the certificates, and whose monitoring-adequacy converse is the design reading. This paper consolidates the principle once, with the exactness class stated on the theorem (finite-horizon finite models and static observation are complete; elsewhere the duals are sound exclusion tests and the residual dynamic gap remains Open Problem 1), the causes kept distinct, and a concordance of every audited count — which integer is which kernel on which system, with each number cited to its source paper and machine-verified here in exact arithmetic (8 check families). Two corrections ship with the concordance: the delay rule on the integer grid is the *non-strict* identity — viability at deadline T_{obs} iff $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$ (the boundary cell $z_0 = 1 + T_{\text{obs}}$ is viable; the strict form fails there) — and the coarsest adequate monitoring is *not* “the” partition into maximal common-action sets (on the audited target the two maximal sets overlap, so no such partition exists; two minimal adequate partitions exist instead). No completed item is modified; the concordance cites.

Purpose

The programme’s papers state one principle many times: viability under incomplete observation is governed by a recursively defined selector, its emptiness is the obstruction, each certificate is a checkable witness of emptiness at one factor, and the design question inverts the test. The principle is latent in the calculus paper’s Theorems 1–7 and Propositions 1–8, in the viable selector’s meta-theorem, and in the catalogue notes’ audited instances. This concordance states it once with its scope, tables the dual witnesses against their sources, states the design converse with two corrections (the delay identity’s boundary case; the coarsest-partition existence), and lists every audited count with its meaning and source — so that each integer is cited once, from one place. Nothing here modifies a completed item.

The selector principle (primal)

Fix a policy class Π , observation structure $\mathcal{I}$, review length Δ , horizon N . For an information state B the recursively viable selector is (Abaee 2026b, Definition 2; Abaee 2026a, Theorem 1)

$$\Gamma_N(B) = \{\alpha : \text{Reach}_{[0, \Delta]}(B, \alpha) \subseteq \mathcal{V} \\ \text{and every successor } B' \in \mathcal{W}_{N-1}\},$$

with $\mathcal{W}_0 = \{B \subseteq \mathcal{V}\}$ and $\mathcal{W}_N = \{B : \Gamma_N(B) \neq \emptyset\}$; the one-step factors nest (Abaee 2026b, Proposition 1), $\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_{\mathcal{V}}^B(B) \subseteq U^B(B)$.

Theorem 1 (exactness class). In a finite-horizon finite model, B admits an N -step observation-based policy in Π keeping every compatible trajectory in $\mathcal{V}$ if and only if $\Gamma_N(B) \neq \emptyset$; under static observation the identity reduces to the existence of an open-loop robust control (Abaee 2026a, Theorems 1 and 6; Abaee 2026b, Theorem 2, with the finite qualifier in the theorem box). Beyond the class, exact completeness requires a non-anticipative measurable selection (Bertsekas – Shreve) and is not claimed.

Theorem 2 (soundness elsewhere). Under the standing C^1 /closed-graph/comparison hypotheses, emptiness of any ladder factor, or of the blind-window predecessor at the observation horizon, implies $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ (Abaee 2026a, Theorems 2–5). The converse in continuous time — a measurable selector from pointwise-nonempty Γ — is Open Problem 1, open.

Dual witnesses

The stochastic reading (deficit dual) replaces $\Gamma_N(B)$ by the belief-state safety value $V_k(b)$: the deterministic degeneration recovers the selector, $\Delta_k(b) = 1 - V_k(b)$ is the dual deficit, and the chance-constrained/min-mass bounds are dual feasible solutions — exact rational α -vectors on rational data (Abaee 2026d). Expected survival is not conflated with worst-case viability: the robust certificates appear only at the degeneration.

The design converse

Corollary 1 (one-step monitoring adequacy). For finite states and a target $K \subseteq \mathcal{V}$, an observation structure admits a one-step viable observation-based policy on K iff every fibre meeting K satisfies $\bigcap_{x \in F \cap K} \mathcal{R}_{\mathcal{V}}(x) \neq \emptyset$; pairwise intersection is necessary and, off convexity, not sufficient (the three two-point codices; Abaee 2026e, E1). Multi-step adequacy does not compose; necessity-only remains the general claim.

Certificate	Factor emptied	Dual witness	Source
Admissibility	$U^B(B) = \emptyset$	no common instrument	Abae 2026a, Prop. 5
Common action	$\mathcal{R}_V^B(B) = \emptyset$	Farkas pair on the stacked rows; Helly $m+1$ if convex	Abae 2026a, Thm. 2, Prop. 4; 2026c, E1–E2
Tube	$\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$	uniform-margin exit within the review	Abae 2026a, Prop. 2
Timing	$\text{Pre}_{\text{blind}}$ empty at T_{obs}	$\sigma^*(B_0) < T_{\text{obs}}$; scalar margin $\inf q/\varepsilon$	Abae 2026a, Thm. 3
Finite-time exit	Γ empty at singletons	Isaacs/Dini outward drift on a strip	Abae 2026a, Thm. 4
Fibre (certification)	no exact label $\ell \circ O = \mathbf{1}_K$	a fibre meeting K and K^c ; certainly-safe set	Abae 2026a, Prop. 7, Cor. 1
Class restriction	$\Gamma_{\Pi} = \emptyset, \Gamma_{\Pi'} \neq \emptyset$	uncorrected law’s positive drift; corrected law’s zero	Abae 2026a, Remark 4; 2026b, Cor. 1

Table 1: The dual witnesses: each certificate is a checkable witness that one factor of Γ is empty. Weak duality always: a witness implies nonviability in the declared class. Strong duality exactly in the class of Theorem 1: emptiness iff a witness exists. Causes are not unified — dynamic exit empties Γ under full information (not epistemic); the fibre row is a certification limit, not a policy-viability obstruction; the class row can fire under injective observation (Abae 2026b, Section 7’s discipline).

Corollary 2 (coarsest adequate monitoring). Adequate partitions form an upper set in the partition lattice, so minimal-cost adequate partitions exist — but they need not be unique, and they are *not* “the” partition into maximal common-action sets: on the audited four-state target the maximal common-action sets $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$ *overlap*, so no such partition exists, and two distinct two-cell adequate partitions are minimal (Abae 2026e, E3, E7; verified here, check C2).

Corollary 3 (four design rules, exact). (1) *Timing*: on the delayed hidden-regime hold class, viability at deadline T_{obs} iff $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$, equivalently $z_0 \geq 1 + T_{\text{obs}}$ — the *non-strict* identity; the boundary cell $z_0 = 1 + T_{\text{obs}}$ is viable (the review lands exactly at exit), so the strict form of the earlier editions is false on the grid (check C1). (2) *Coarseness*: a fibre merging states with disjoint safe-action sets obstructs at that information state even when every branch is viable; remedy: split the fibre. (3) *Aggregation/certification*: an index whose fibres cross a floor cannot support an exact safe/unsafe verdict; report the certainly-safe set or fall silent — label selection, not control. (4) *Bias*: the uncorrected certainty-equivalence law can empty the selector of a system whose corrected-law selector is nonempty; the $\frac{21}{100}$ figure is the *uncorrected* canonical law’s drift, the corrected law’s drift being identically zero (check C6).

Restorative moves

For a nonviable B in the exactness class, the minimal restorative is one of: a fibre split restoring a nonempty common safe action; an instrument added to $U^B(B)$; a shortened T_{obs} ; a bias correction. On a two-point belief the only nontrivial split is into the two singletons — the audited distances are immediate corollaries of Theorem 1 and Corollary 1 (Abae 2026f; the move families, not the arithmetic, are the diagnostic).

The count concordance

All of the following are re-verified in exact arithmetic by the record (checks C1–C8). *Two-patch working system* (Abae 2026a, Section 8: patches $i \in \{1, 2\}$, stocks $z_i \in \{0, 1, 2, 3\}$, regeneration $+1$ cap 3, unprotected-patch damage 2, instruments $\{1, 2\}$, floors $z_i \geq 1$): **36** = the pair family, $\binom{9}{2}$ two-state beliefs of the nine floor-safe states; **28** = the full-information kernel, $\binom{8}{2}$, the pairs avoiding the corner $(1, 1)$, whose singleton is nonviable (physical failure); **26** = the aggregate per-cell kernel; **24** = the institutional (aggregate, no-repeat codex) kernel; **25** = the full-information codex kernel — the four form the Boolean product with incomparable middles 26, 25 and institutional meet (Abae 2026f, E7–E8). **12/16** = the decentralized unanimity kernel and the deadlock shadow under split observation (Abae 2026g); the **16** anti-symmetric-sustaining law pairs for

the audited belief are a numerical coincidence with the lost pairs, no identity; **256** = $2^4 \times 2^4$ law pairs. *Mode uncertainty* (Abae 2026h): frozen regimes $36 = 36$; the mode-blind **28** is the *setwise coincidence* with the base full-information kernel under different damage routing, generated entirely by the mode-blind death of the corner. *Monitoring* (Abae 2026e): **48** = the hidden-regime grid ($z_0 \in \{1.0, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$); **7 of 15** = admissible partitions of the four-state target, two minimal two-cell ones. *Conventions*: σ^* on the hold class is the discrete survival supremum $\lfloor z_0 - 1 \rfloor$ (continuous successor value $z_0 - 1$); the CE drift $\geq \frac{21}{100}$ on $[1, 2]$ belongs to the canonical *uncorrected* law, the corrected law’s drift being identically zero. *Continuous benchmark* (Abae 2026i): critical aggregate $Y^* = \frac{27}{5}$ for the two-stock model with the single demand-floor control set; the $Y = 6$ fibre obstructed with the two-point dual measure of margin $\frac{3}{50}$; fibre width $\Delta x_1(Y) = Y - 4$.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `paper2_selector_concordance_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all eight check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review, under the programme’s verification discipline.